Academic Year 2026/27

Cardiff Metropolitan University | Prifysgol Metropolitan Caerdydd

School of Technologies

Assessment Brief

Module Code	Module Title
DAT6007	Cloud Computing Deployment

Academic Year	Semester
2026/27	1

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Assessment Details

Assessment title	Abr.	Weighting
CloudCart – Scalable E-Commerce Platform	WRIT1	75%

Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.

Task/Assessment brief:

The student will produce a 3000-word report on the design, deployment, and evaluation of the CloudCart platform. The report should discuss the cloud technologies used, security and scalability features, testing, and key technical decisions, supported by relevant evidence where appropriate. The details of the CloudCart project are given below.

CloudCart – Scalable E-Commerce Platform

The rapid growth of e-commerce solutions has increased the need for secure and scalable platforms that can handle high volumes of user traffic and product data. Cloud computing has become a vital tool for hosting online retail applications, enabling remote access, real-time order processing, and high scalability. With rising concerns around cybersecurity and data privacy, it is essential to ensure that applications comply with regulations such as GDPR while protecting customer data through encryption, secure authentication, and virtual payments.

In this project, students will deploy an e-commerce application, CloudCart, on a cloud platform like AWS, Azure, or Google Cloud. The focus will be on implementing security measures, configuring scalable infrastructure, and ensuring reliable performance during high-traffic events. The goal is to help students understand the technical and regulatory aspects of deploying secure, scalable e-commerce applications in the cloud. The main features of CloudCart are given below:

Key Features of CloudCart:

1. **Product Catalog Management** – Add, edit, and organize products with real-time stock updates.
2. **Shopping Cart and Checkout** – Smooth cart experience with **virtual payments** to simulate transactions.
3. **User Authentication** – Secure login/sign-up with multi-factor authentication and OAuth support.
4. **Order Tracking** – Real-time updates on orders and delivery status.
5. **Scalability Features** – Auto-scaling, CDN-backed content delivery, and load balancing to handle traffic spikes.
6. **Wishlist and Favourites** – Allows customers to save products for future purchase, enhancing engagement.
7. **Virtual Payments Management** – Simulated payment system for handling multiple virtual payment methods and validating transactions in a safe environment.

Deliverables:

1. *A link to the deployed CloudCart application.*
2. *A **3000-word report** covering the application design, cloud deployment, security implementation, scalability features, testing, evaluation, and justification of technical decisions.*

3. *A ZIP file containing the complete source code and supporting files.*

Word count (or equivalent):	3000

This a reflection of the effort required for the assessment. Word counts will normally include any text, tables, calculations, figures, subtitles and citations. Reference lists and contents of appendices are excluded from the word count. Contents of appendices are not usually considered when determining your final assessment grade.

Academic or technical terms explained:

- **Cloud Computing / Cloud Platform** – The foundational technology for deployment.
- **AWS, Azure, Google Cloud** – The specific cloud providers suggested for the project.
- **Deploy / Deployment –** The act of installing, configuring, and running the application on the cloud servers.
- **Configure –** Setting up the various cloud services and application components**.**
- **Infrastructure –** The underlying cloud resources (servers, databases, networks).

Artificial Intelligence Models – Guidance for this assessment:

Artificial Intelligence (AI) models can be a powerful tool to support your learning. The University has provided some resources to support you in its appropriate usage:

- AI Student Hub
- Can I Use AI in my Assessment?
- Student Code of Conduct for AI Use
- Cite Them Right resource on citing materials relating to AI (if permitted)

As per the academic regulations (Academic Handbook Ah1_08), in all cases you must submit work that is your own, acknowledging any part of it that has been informed by another source – including that which is AI generated. Upon submission of work, you will be asked to confirm the following statement:

> *I confirm that this assignment is my own work, except where I have acknowledged the use of works from other sources, including the use of any artificial intelligence (AI) tools, in accordance with what is allowable as described in the assessment brief.*

Please note the following:

- AI should not be used as a substitute for your own knowledge, and you should never include any material that you do not understand and could not explain if asked.
- Not being able to explain your work when asked is likely to be a key factor when considering cases of academic misconduct related to AI.

The following information provides specific guidance for this assessment about what level of AI use is appropriate for this assessment. Remember that in all cases you must submit work that is your own, acknowledging any part of it that has been provided by another source.

NO USE OF GENERATIVE AI EXPECTED • Your assignment should be produced using information sourced by you from your learning materials and academic sources and cited appropriately. • AI tools for checking spelling, grammar and referencing may be used.	
AI ACKNOWLEDGED • You can use AI tools to learn about your topic, as part of your study, or in preparing initial guidance on assignments (e.g. headline structure, suggestions for inclusion of topics). • Any materials that you have sourced from AI should be rewritten or reconfigured and integrated into your own work and referenced appropriately. It is recommended that this is confirmed by a relevant academic source. • Any support gained from AI should be acknowledged in a statement at the end of the assignment, making clear what the support was, and how you used it and developed it for your own work. Example statements are available in the Student Code of Conduct for AI Use	☑
AI EMBEDDED • Use of AI is an integral and expected part of the assessment. • The explicit inclusion of AI within the assessment means that instructions on the expected use will be part of the assessment brief. • Your assessment brief will describe how you should acknowledge the way in which you used AI tools.	

Submission Details

Submission Deadline:	This will be provided on the Moodle submission point.
Submission Time:	By 4.00pm on the deadline day.
Estimated Feedback Return Date	This will normally be 20 working days after initial submission.
File Format:	The assessment must be submitted as a pdf document (save the document as a pdf in your software) and submit through the Turnitin submission point in Moodle. **Your assessment should be titled with your:** **student ID number, module code and assessment ID, e.g. st12345678 BHL5007 WRIT1**
Late Submission Window Eligibility	Where submissions are eligible for the late-submission window this will be communicated in the relevant assessment submission point within Moodle.
Feedback	Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the guidance on how to access your feedback. All marks are provisional and are subject to quality assurance processes and confirmation at the programme Examination Board.

Assessment Criteria

Learning outcomes assessed

Learning Outcomes	Assessment Criteria
[LO1] Evaluate and explain cloud computing technology. [LO2] Explore scalability and cloud architecture. [LO3] Contrast and compare Simple Object Access Protocol (SOAP) and Representational State Transfer (REST) Web Services. [LO4] Develop multi-tier web service applications hosted on an industry standard PaaS cloud platform. [LO5] Evaluate cloud containers design and concept.	[AC1]: Code deployment on cloud (30%): [LO4] [AC2]: Effective use of Cloud Resources (20%): [LO1], [LO4] [AC4] Report (50%): [LO1], [LO2], [LO3], [LO4], [LO5].

Other skills/attributes developed

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be

highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

ETHICAL –The data generated by the end user devices, and data collected by the systems (Clouds) (e.g., personal data) should be handled, saved and processed with proper care adhering to international and regional laws and regulations such as GDPR. This module particularly focuses on how the privacy could be preserved using distributed machine learning.

DIGITAL – Programming and research skills accumulated during the course and through the assignment will enhance student's hands on research skills, use of IDEs, virtual machines, and real hardware devices.

GLOBAL – Universally accepted research skills, research ideas implementation whether simulation or real hardware will be utilised in the assignment. Skills gained through this are location, language agnostic and are applicable across the globe.

ENTREPRENEURIAL – Inclusive pedagogy will be considered which will be based on the interest of the students in the topic. Moreover, bringing the classroom concepts to practical implementation will be considered in the sessions so that students can realize how classroom content can be used in real life settings and how they can convert their ideas into a product.

Marking/Assessment Criteria

The higher-level breakdown of the marks is as:

Component	Marks
Code Deployment on Cloud	30
Effective use of Cloud Resources	20
Report	50
Total	100

The detailed breakdown of the marks is provided on next page.

Marking/Assessment Criteria

Marks	Criteria	< 35% (Fail)	35 - 39% (Narrow Fail)	40- 49% (3rd)	50-59% (2:2)	60-69% (2:1)	70-79 % (1st)	80-100% (1st Class +)
Code Deployment on Cloud (30%)								
15	Deployment & Core Features	Application not deployed or completely non-functional. Fewer than 3 features implemented.	Partially deployed or extremely unstable. 1–2 user journeys work intermittently; major parts missing or broken.	Application is deployed but unstable. 3–4 features are implemented but major bugs prevent key user journeys.	Application is deployed and mostly stable. 5–6 key features are implemented with only minor bugs.	Application is fully deployed and stable. All 7 key features implemented; only minor polish required.	Application is fully deployed, exceptionally stable, highly available; all features implemented correctly, polished UX and monitoring in place.	Deployment demonstrates technical excellence: fully automated CI/CD pipeline, zero-downtime updates, resilience testing, proactive monitoring/alerting, and clear evidence of scalability under real-world loads.
15	Code Security	Shows serious negligence or critical security vulnerabilities (e.g., secrets exposed, SQLi/XSS unmitigated).	Multiple significant security flaws present; basic protections missing or applied inconsistently (e.g., some plaintext secrets, weak password handling).	Security is flawed (e.g., plain text passwords, no HTTPS).	Demonstrates basic security: passwords are hashed but may lack consistent HTTPS or input validation.	Demonstrates robust security: passwords hashed, HTTPS/SSL enforced, input validation used and basic logging.	Exemplary security: strong auth/authorization (RBAC/OAuth), encryption at rest, secrets management, secure headers, logging/monitoring and regular vulnerability scanning; OWASP mitigations evident.	Security is industry-leading: comprehensive secure SDLC practices, penetration testing evidence, automated vulnerability scanning in CI/CD, compliance with multiple frameworks (e.g., ISO, NIST), and documented incident response plan.
Effective use of Cloud Resources (20%)								
10	Scalability & Architecture	Shows no understanding of cloud resources or	Little/no scalability: ad-hoc or single-instance setup,	Uses only static, non-scalable resources	Uses either auto-scaling or a load balancer. CDN may be present but not optimized.	Effectively uses auto-scaling, load balancing, and a CDN.	Advanced architecture: infra-as-code, multi-region or fault-tolerant design, caching	Architecture is multi-cloud or hybrid setup, automated failover, blue/green or canary deployments, proactive

		scalable architecture.	no redundancy or monitoring.	(e.g., a single VM).		Architecture designed for performance.	layers, observability (metrics/tracing/logs) and clear cost/performance tuning.	cost optimization, and advanced observability (tracing, distributed monitoring, predictive autoscaling)
10	Service Integration	Minimal or ineffective use of cloud services.	Basic or misconfigured integration of services; manual operational workarounds present.	Primarily uses basic IaaS (Virtual Machines) for everything.	Uses a mix of IaaS and PaaS services.	Strategic use of managed services to reduce overhead and improve reliability.	Extensive, well-architected use of cloud-native/managed services (serverless, managed DBs, messaging, secret manager, CDNs), secure integrations and automated provisioning.	Integration demonstrates mastery: advanced orchestration of diverse services, event-driven/serverless-first design, seamless interoperability, cost/security optimized integrations, with automated disaster recovery setup.
Report (50%)								
25	System Architecture, including UML Diagrams & Technical Description	No description. No diagram. Explanations are minimal, incorrect, or missing.	Attempted diagrams/descriptions but incomplete or incorrect; many major components missing; little technical detail.	Diagrams are missing or rudimentary. Descriptions lack technical detail.	Includes diagrams and describes most components, but justifications are vague.	Includes clear, professional diagrams. Thoroughly explains and justifies all components.	professional UML (component/sequence/deployment) and architecture diagrams, detailed technical descriptions, clear rationale, trade-offs, data flows and performance/security considerations.	Outstanding report: comprehensive UML set (component/sequence/deployment/state), advanced justifications with alternatives considered, evaluation of trade-offs, and exceptional clarity suitable for publication or professional documentation.
10	Security & Compliance Analysis	No discussion of security or compliance.	Very brief or superficial mention of security with no concrete controls; compliance only	Offers only a brief, vague mention of security. No compliance discussion.	Lists security measures but lacks depth. Mentions GDPR without specific steps.	Detailed analysis of security measures and specific steps taken for GDPR/compliance.	Comprehensive security & compliance analysis: threat modeling, risk assessment, implemented controls, incident response plan, data	Gold-standard security & compliance: detailed mapping to multiple frameworks (e.g., GDPR, HIPAA, ISO27001), evidence of compliance processes, auditing, governance policies, risk

			referenced in passing.				mapping and explicit mapping to GDPR/other standards.	registers, and proactive improvement plan.
10	Scalability Explanation & Reflection	No discussion of scalability or reflection on the project.	Very limited reflection; claims scalability without specifics or measurable evidence.	States that the app is scalable but provides no explanation.	Mentions scaling services but lacks detail on configuration.	Explains the scaling strategy in detail. Includes thoughtful retrospective.	Insightful reflection with benchmarks/load-test results, concrete scaling configs, lessons learned, and a clear roadmap for further improvements.	Exceptional reflection: critical evaluation of scaling trade-offs, empirical benchmarking with graphs/metrics, cost-performance analysis, and visionary roadmap anticipating future trends (AI ops, predictive scaling).
5	References	No references are provided, or they are completely irrelevant to the project.	Some references provided (few), but poorly formatted, not credible or lacking diversity.	6–10 relevant references are included from credible sources with acceptable formatting.	6–10 relevant references from credible sources, well formatted and clearly tied to sections of the report.	>10 relevant references included from credible, diverse sources with correct formatting and relevance.	Extensive (>15) high-quality references (official docs, standards, peer-reviewed or authoritative sources), properly formatted and cited inline where used.	References are exemplary: >20 sound and diverse sources (academic, industry standards, white papers, documentation), meticulously formatted, and critically engaged with in the text.

Further Information

Who can answer questions about my assessment?

Questions about the assessment should be directed to the staff member who has set the task/assessment brief. This will usually be the Module Leader. They will be happy to answer any queries you have.

Staff members can often provide feedback on an assignment plan but cannot review any drafts of your work prior to submission. The only exception to this rule is for Dissertation Supervisors to provide feedback on a draft of your dissertation.

Referencing and independent learning

Please ensure you reference a range of credible sources, with due attention to the academic literature in the area. The time spent on research and reading from good quality sources will be reflected in the quality of your submitted work.

Remember that what you get out of university depends on what you put in. Your teaching sessions typically represent between 10% and 30% of the time you are expected to study for your degree. A 20-credit module represents 200 hours of study time. The rest of your time should be taken up by self-directed study.

Unless stated otherwise you must use the **HARVARD** referencing system. Further guidance on referencing can be found in the Study Smart area on Moodle and using Cite Them Right (use your university login details to access the site). Correct referencing is an easy way to improve your marks and essential in achieving higher grades on most assessments.

Technical submission problems

It is strongly advised that you submit your work at least 24 hours before the deadline to allow time to resolve any last minute problems you might have. If you are having issues with IT or Turnitin you should contact the IT Helpdesk on (+44) 2920 417000. You may require evidence of the Helpdesk call if you are trying to demonstrate that a fault with Moodle or Turnitin was the cause of a late submission.

Late submission and mitigating circumstances

If you are experiencing personal difficulties which are impacting your ability to engage with assessment, there are a range of support options available, including via your Personal Academic Tutor, the Mitigating Circumstances procedure and the Support to Study procedure.

The Mitigating Circumstances policy and procedure sets out when and how you can:

- Use a late submission window to submit eligible assessments up to 5 working days after the deadline. Student can use up to four late submission permits per academic level to allow work submitted in the late submission window to be recorded as 'on time'.
- Defer two assessments per level of study to the next submission opportunity through self-declaration of exceptional circumstances.

Where more assessment adjustments are required, you can use the Support to Study procedure.

Here's a link to MetCentral where there is more detailed information on the support offered through Mitigating Circumstances and Support to Study and how to access it.

Students with a disability or chronic illness should contact the University's Student Wellbeing Team to discuss any appropriate reasonable adjustments which may be made to their learning and assessments.

Academic Misconduct

Cardiff Met takes issues of academic misconduct **extremely seriously.** The University has procedures and penalties for dealing with academic misconduct. These are explained in full in the University's Academic Misconduct regulations and procedures under Volume 1, Section 8 of the Academic Handbook. The Module Leader reserves the right to interview students regarding any aspect of their work submitted for assessment.

Types of Academic Misconduct, include:

Plagiarism, which can be defined as using without acknowledgement another person's words or ideas and submitting them for assessment as though it were one's own work, for instance by copying, translating from one language to another or unacknowledged paraphrasing. Further examples include:

- Use of any quotation(s) from the published or unpublished work of other persons, whether published in textbooks, articles, the Web, or in any other format, where quotations have not been clearly identified as such by being placed in quotation marks and acknowledged.
- Use of another person's words or ideas that have been slightly changed or paraphrased to make it look different from the original.
- Summarising another person's ideas, judgments, diagrams, figures, or computer programmes without reference to that person in the text and the source in a bibliography/reference list.
- Use of assessment writing services, essay banks and/or any other similar agencies (NB. Students are commonly being blackmailed after using essay mills).
- Use of unacknowledged material downloaded from the Internet.
- Re-use of one's own material except as authorised by your degree programme.

Collusion, which can be defined as when work that that has been undertaken with others is submitted and passed off as solely the work of one person. Modules will clearly identify where joint preparation and joint submission are permitted, in all other cases they are not.

Fabrication of data, making false claims to have carried out experiments, observations, interviews or other forms of data collection and analysis, or acting dishonestly in any other way.

How is my work graded?

Assessment grading is subject to thorough quality control processes. You can view a summary of these processes on the Assessment Explained Infographic.

Grading of work at each level of Cardiff Met degree courses is benchmarked against a set of general requirements set out in Volume 1, Section 4 of our Academic Handbook. A simplified version of these Grade Band Descriptors (GBDs) with short videos explaining some of the academic terminology used can be accessed for Foundation, 1st year, 2nd year and 3rd year undergraduate and MSc programmes.

We would strongly recommend looking at the Study Smart area of Moodle to find out more about assessments and key academic skills which can have a significant impact on your grades. Always check your work thoroughly before submission.

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